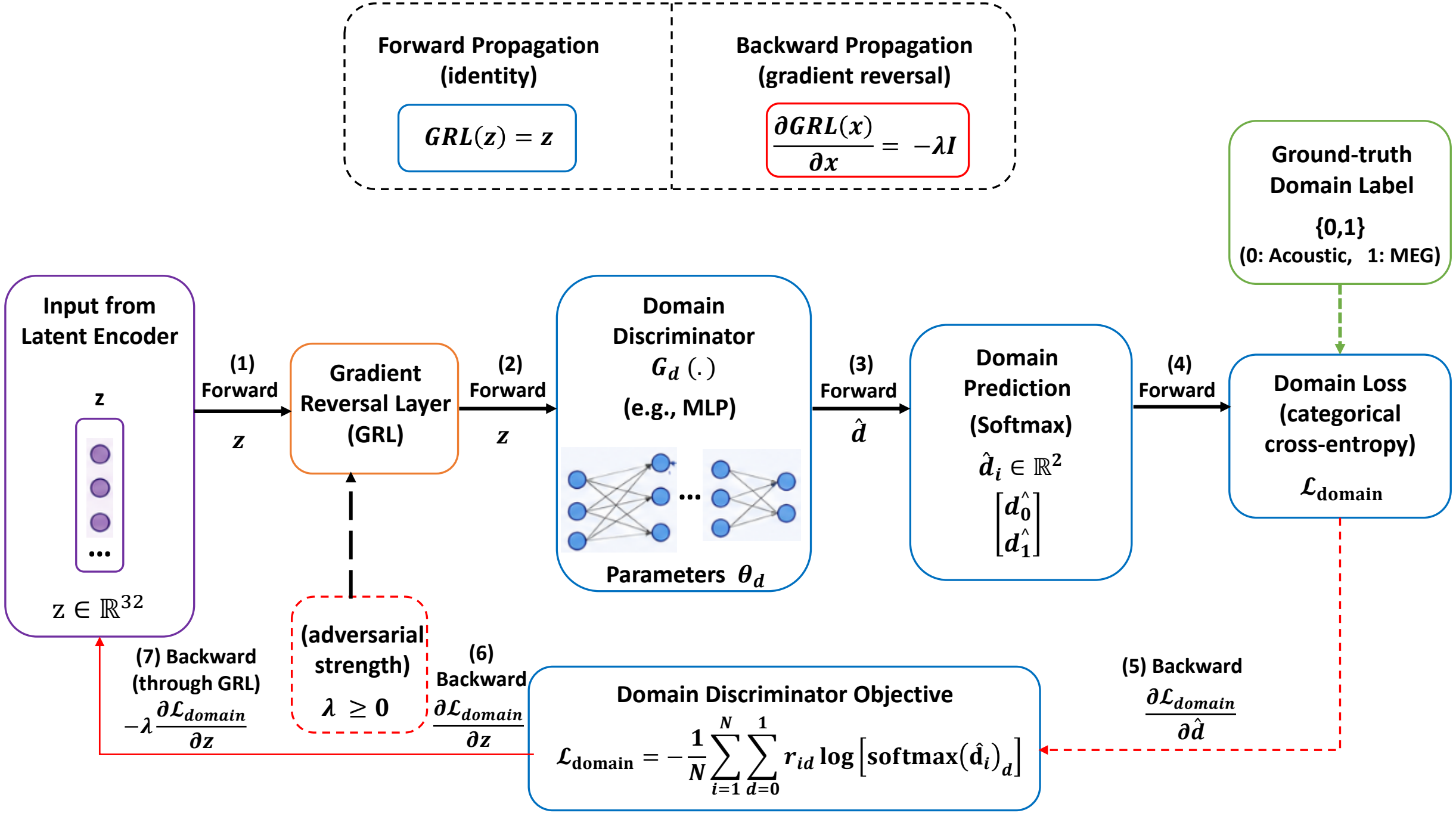

Forward Propagation
(identity)
$GRL(z) = z$
Backward Propagation
(gradient reversal)
$\frac{\partial GRL(x)}{\partial x} = -\lambda I$
Ground-truth
Domain Label
{0,1}
(0: Acoustic, 1: MEG)
Input from
Latent Encoder
z
$z \in \mathbb{R}^{32}$
(1)
Forward
z
Gradient
Reversal Layer
(GRL)
(2)
Forward
z
Domain
Discriminator
$G_d(.)$
(e.g., MLP)
Parameters θ_d
(3)
Forward
$\hat{d}$
Domain
Prediction
(Softmax)
$\hat{d}_i \in \mathbb{R}^2$
$\begin{bmatrix} \hat{d_0} \\ \hat{d_1} \end{bmatrix}$
(4)
Forward
Domain Loss
(categorical
cross-entropy)
$\mathcal{L}_{domain}$
(adversarial
strength)
$\lambda \geq 0$
(7) Backward
(through GRL)
$-\lambda \frac{\partial \mathcal{L}_{domain}}{\partial z}$
(6)
Backward
$\frac{\partial \mathcal{L}_{domain}}{\partial z}$
Domain Discriminator Objective
$\mathcal{L}_{domain} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{d=0}^{1} r_{id} \log\left[\text{softmax}(\hat{d}_i)_d\right]$
(5) Backward
$\frac{\partial \mathcal{L}_{domain}}{\partial \hat{d}}$